

Zone Sizing Summary for UNIT KING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

05/29/2026
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Air System Information

Air System Name **UNIT KING**
Equipment Class **TERM**
Air System Type **PKG-FC**

Number of zones **1**
Floor Area **353.3** sqft
Location **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone CFM Sizing **Sum of space airflow rates**
Space CFM Sizing **Individual peak space loads**

Terminal Unit Sizing Data - Cooling

Zone Name	Total Coil Load (MBH)	Sens Coil Load (MBH)	Coil Entering DB / WB (F)	Coil Leaving DB / WB (F)	Water Flow @ 10.0 F (gpm)	Time of Peak Coil Load	Zone CFM/sqft
UNIT KING	6.7	6.4	73.4 / 56.6	56.7 / 49.7	-	July 19:00	1.10

Terminal Unit Sizing Data - Heating, Fan, Ventilation

Zone Name	Heating Coil Load (MBH)	Heating Coil Ent/Lvg DB (F)	Htg Coil Water Flow @20.0 F (gpm)	Fan Design Airflow (CFM)	Fan Motor (BHP)	Fan Motor (kW)	OA Vent Design Airflow (CFM)
UNIT KING	3.3	72.0 / 80.6	-	390	0.029	0.023	39

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
UNIT KING	5.7	June 19:00	2.6	353.3

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
UNIT KING						
UNIT 533-KING	5.7	June 19:00	390	2.6	353.3	1.10

Ventilation Sizing Summary for UNIT KING

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1. Summary

Ventilation Sizing Method **Sum of Space OA Airflows**
Design Ventilation Airflow Rate **39** CFM

2. Space Ventilation Analysis

2.1 Zone: UNIT KING

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
UNIT KING										
UNIT 533-KING	353.3	3.5	390.1	5.00	0.06	0.0	0.0	0.00	38.9	0.0
Totals			390.1						38.9	0.0

Air System Heat Balance Summary for UNIT KING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
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Table 1. System Loads

COMPONENT LOADS	DESIGN COOLING - JULY 19:00			DESIGN HEATING		
	OA DB / WB 100.1 F / 64.3 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	6263	426	-	3255	0
Plenum Load	-	0	0	-	0	0
Exhaust Fan Load	39 CFM	0	-	39 CFM	0	-
Ventilation Load	39 CFM	1006	-138	39 CFM	1953	0
Ventilation Fan Load	39 CFM	58	-	39 CFM	-58	-
Zone Fan Coil Fans Load	-	78	-	-	-78	-
>> Total System Loads	-	7405	288	-	5073	0
Cooling Coil	-	1001	0	-	0	0
Heating Coil	-	0	-	-	1828	-
Terminal Unit Cooling	-	6414	324	-	0	0
Terminal Unit Heating	-	0	-	-	3253	-
>> Total Conditioning	-	7415	324	-	5081	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

Zone Heat Balance Component	DESIGN COOLING - JULY 19:00			DESIGN HEATING		
	OA DB / WB 100.1 F / 64.3 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	294 sqft	1439	-	294 sqft	792	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	36 sqft	304	-	36 sqft	319	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	353 sqft	599	-	353 sqft	355	-
Interior Wall Convection	330 sqft	483	-	330 sqft	230	-
Ceiling Convection	353 sqft	1516	-	353 sqft	942	-
Overhead Lighting Convection	98 W	148	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	353 W	904	-	0 W	0	-
People Convection	4	244	424	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	-	5637	424	-	2638	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Zone Sizing Summary for UNIT QQ

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
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Air System Information

Air System Name **UNIT QQ** Number of zones **1**
Equipment Class **TERM** Floor Area **454.7** sqft
Air System Type **PKG-FC** Location **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months **Jan to Dec** Zone CFM Sizing **Sum of space airflow rates**
Sizing Data **Calculated** Space CFM Sizing **Individual peak space loads**

Terminal Unit Sizing Data - Cooling

Zone Name	Total Coil Load (MBH)	Sens Coil Load (MBH)	Coil Entering DB / WB (F)	Coil Leaving DB / WB (F)	Water Flow @ 10.0 F (gpm)	Time of Peak Coil Load	Zone CFM/sqft
UNIT QQ	8.3	8.0	73.4 / 56.6	57.4 / 50.2	-	August 14:00	1.11

Terminal Unit Sizing Data - Heating, Fan, Ventilation

Zone Name	Heating Coil Load (MBH)	Heating Coil Ent/Lvg DB (F)	Htg Coil Water Flow @20.0 F (gpm)	Fan Design Airflow (CFM)	Fan Motor (BHP)	Fan Motor (kW)	OA Vent Design Airflow (CFM)
UNIT QQ	3.8	72.0 / 79.8	-	506	0.037	0.030	50

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
UNIT QQ	7.4	September 14:00	3.1	454.7

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
UNIT QQ						
UNIT 516-QQ	7.4	September 14:00	506	3.1	454.7	1.11

Ventilation Sizing Summary for UNIT QQ

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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1. Summary

Ventilation Sizing Method **Sum of Space OA Airflows**
Design Ventilation Airflow Rate **50** CFM

2. Space Ventilation Analysis

2.1 Zone: UNIT QQ

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
UNIT QQ										
UNIT 516-QQ	454.7	4.5	506.1	5.00	0.06	0.0	0.0	0.00	50.0	0.0
Totals			506.1						50.0	0.0

Air System Heat Balance Summary for UNIT QQ

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Table 1. System Loads

COMPONENT LOADS	DESIGN COOLING - AUGUST 14:00			DESIGN HEATING		
	OA DB / WB 104.9 F / 65.8 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	7783	511	-	3843	0
Plenum Load	-	0	0	-	0	0
Exhaust Fan Load	50 CFM	0	-	50 CFM	0	-
Ventilation Load	50 CFM	1552	-189	50 CFM	2514	0
Ventilation Fan Load	50 CFM	74	-	50 CFM	-74	-
Zone Fan Coil Fans Load	-	101	-	-	-101	-
>> Total System Loads	-	9510	322	-	6181	0
Cooling Coil	-	1549	0	-	0	0
Heating Coil	-	0	-	-	2353	-
Terminal Unit Cooling	-	7979	313	-	0	0
Terminal Unit Heating	-	0	-	-	3839	-
>> Total Conditioning	-	9528	313	-	6192	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

Zone Heat Balance Component	DESIGN COOLING - AUGUST 14:00			DESIGN HEATING		
	OA DB / WB 104.9 F / 65.8 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	351 sqft	1524	-	351 sqft	917	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	36 sqft	545	-	36 sqft	313	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	455 sqft	942	-	455 sqft	418	-
Interior Wall Convection	387 sqft	614	-	387 sqft	252	-
Ceiling Convection	455 sqft	1883	-	455 sqft	1156	-
Overhead Lighting Convection	126 W	191	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	455 W	1164	-	0 W	0	-
People Convection	5	314	546	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	-	7177	546	-	3056	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Zone Sizing Summary for UNIT QQ XXL

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
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Air System Information

Air System Name **UNIT QQ XXL** Number of zones **1**
Equipment Class **TERM** Floor Area **444.8** sqft
Air System Type **PKG-FC** Location **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months **Jan to Dec** Zone CFM Sizing **Sum of space airflow rates**
Sizing Data **Calculated** Space CFM Sizing **Individual peak space loads**

Terminal Unit Sizing Data - Cooling

Zone Name	Total Coil Load (MBH)	Sens Coil Load (MBH)	Coil Entering DB / WB (F)	Coil Leaving DB / WB (F)	Water Flow @ 10.0 F (gpm)	Time of Peak Coil Load	Zone CFM/sqft
UNIT QQ XXL	8.1	7.9	73.4 / 55.9	57.1 / 49.3	-	September 14:00	1.11

Terminal Unit Sizing Data - Heating, Fan, Ventilation

Zone Name	Heating Coil Load (MBH)	Heating Coil Ent/Lvg DB (F)	Htg Coil Water Flow @20.0 F (gpm)	Fan Design Airflow (CFM)	Fan Motor (BHP)	Fan Motor (kW)	OA Vent Design Airflow (CFM)
UNIT QQ XXL	3.7	72.0 / 79.7	-	493	0.036	0.029	49

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
UNIT QQ XXL	7.2	September 14:00	2.9	444.8

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
UNIT QQ XXL						
UNIT 500 - QQ XXL	7.2	September 14:00	493	2.9	444.8	1.11

Ventilation Sizing Summary for UNIT QQ XXL

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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1. Summary

Ventilation Sizing Method Sum of Space OA Airflows
Design Ventilation Airflow Rate 49 CFM

2. Space Ventilation Analysis

2.1 Zone: UNIT QQ XXL

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
UNIT QQ XXL										
UNIT 500 - QQ XXL	444.8	4.4	492.9	5.00	0.06	0.0	0.0	0.00	48.9	0.0
Totals			492.9						48.9	0.0

Air System Heat Balance Summary for UNIT QQ XXL

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Table 1. System Loads

COMPONENT LOADS	DESIGN COOLING - SEPTEMBER 14:00			DESIGN HEATING		
	OA DB / WB 99.7 F / 62.8 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	7701	500	-	3677	0
Plenum Load	-	0	0	-	0	0
Exhaust Fan Load	49 CFM	0	-	49 CFM	0	-
Ventilation Load	49 CFM	1271	-318	49 CFM	2459	0
Ventilation Fan Load	49 CFM	73	-	49 CFM	-73	-
Zone Fan Coil Fans Load	-	99	-	-	-99	-
>> Total System Loads	-	9143	182	-	5965	0
Cooling Coil	-	1263	0	-	0	0
Heating Coil	-	0	-	-	2302	-
Terminal Unit Cooling	-	7892	170	-	0	0
Terminal Unit Heating	-	0	-	-	3674	-
>> Total Conditioning	-	9155	170	-	5976	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

Zone Heat Balance Component	DESIGN COOLING - SEPTEMBER 14:00			DESIGN HEATING		
	OA DB / WB 99.7 F / 62.8 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	312 sqft	1394	-	312 sqft	815	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	36 sqft	617	-	36 sqft	313	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	445 sqft	1102	-	445 sqft	408	-
Interior Wall Convection	348 sqft	609	-	348 sqft	226	-
Ceiling Convection	445 sqft	1821	-	445 sqft	1115	-
Overhead Lighting Convection	123 W	187	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	445 W	1138	-	0 W	0	-
People Convection	4	307	534	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	-	7174	534	-	2876	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.